

Harshit Vanya

AI / ML Engineer

+91-9755106375 | harshitvanyagmer@gmail.com | Bhopal, India | 1+ Year Experience

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience designing, building, and deploying end-to-end AI-powered systems including LLM-based agentic workflows, RAG pipelines, and production-grade ML solutions across AWS, Azure, Snowflake, and Databricks. Proven track record delivering config-driven AI/ML pipelines with CI/CD automation, leveraging Python, PySpark, LangChain, LlamaIndex, and vector databases. Recognized for leadership as Data Team Lead, rapidly prototyping AI-native solutions in Agile environments.

KEY SKILLS

AI/ML & GenAI: LLMs | RAG Pipelines | Agent Frameworks (LangChain, LlamaIndex, LangGraph, AutoGen) | Prompt Engineering
Transformer Models | Embeddings | Fine-Tuning | Vector DBs (Pinecone, pgvector, ChromaDB) | Hugging Face | OpenAI API | Azure OpenAI
Data & Cloud: PySpark | Python | SQL | Kafka | Airflow | Snowflake | Databricks | AWS (Lambda, SQS, S3, Bedrock) | Azure (AKS, DevOps, OpenAI) | Docker | CI/CD | Kubernetes | MLflow | Model Drift Detection | Hallucination Detection | Custom Eval Harnesses

WORK EXPERIENCE

Associate Data Engineer / AI-ML Engineer

Mar 2026 - Jun 2026

CloudNexus DigiWeb Services Pvt. Ltd.

- Designed and deployed end-to-end metadata management tool with AI-powered data cataloging for GEICO enterprise client, automating pipeline governance with CI/CD on Azure Kubernetes Service (AKS), reducing manual cataloging effort by 70%.
- Built RAG-based intelligent document retrieval system using LangChain, Azure OpenAI embeddings, and pgvector, achieving 92% retrieval accuracy across 50K+ enterprise documents with sub-2s query latency.
- Developed LLM-powered agentic workflow for automated data quality assessment using LangGraph, reducing validation turnaround from 4 hours to 15 minutes per pipeline run.
- Refactored Microsoft Fabric data pipeline for Canadian Cancer Society, improving modularity by 40% and reducing failures by 60%.
- Implemented config-driven ingestion with AI-assisted anomaly detection, reducing manual intervention by 85%, scaling to 10M+ records/day.
- Promoted to Data Team Lead - coordinated delivery for 5 engineers, code reviews, and mentorship in Agile sprints.
- Built custom LLM evaluation frameworks including hallucination detection and factual grounding, achieving 95% alignment with human evaluators.
- Researched and prototyped emerging frameworks (LangChain, LlamaIndex, AutoGen) for continuous AI engineering improvement.

AI/ML Intern

Sep 2025 - Mar 2026

Cloud Nexus

- Supported LLM integration, model development, and pipeline implementation across 8+ production AI/ML projects.
- Improved ML model accuracy by 18% through hyperparameter tuning and feature engineering on real-world applications.
- Built transformer-based NLP models for text classification and entity extraction with 89% F1 score on production data.
- Implemented vector database solutions (Pinecone, ChromaDB) for semantic search handling 100K+ document embeddings.

PROJECTS

Enterprise RAG Pipeline with Agentic Retrieval	4 Months
· Built end-to-end RAG pipeline with multi-agent orchestration using LangChain and LangGraph for enterprise knowledge management. · Implemented hybrid retrieval (dense + sparse) with Pinecone vector DB, achieving 94% recall@10, 40% improvement over BM25 baseline. · Designed agent planning loops with tool calling, memory modules, and multi-step reasoning for complex query resolution. · Integrated Cohere Rerank and freshness pipelines, reducing hallucination rate by 65%, improving factual accuracy to 91%. · Deployed on AWS (Lambda, API Gateway, SQS) for event-driven processing, handling 5K+ queries/day with p99 latency < 3s.	
Real-Time WiFi/Hotspot Network Monitoring Platform	5 Months
· Built real-time monitoring capturing device info and domains via TShark packet analysis, processing 1M+ packets/hour. · Streamed logs through Apache Kafka (3 partitions, 10K msgs/sec) into Databricks Delta tables for analytics. · Delivered interactive Streamlit dashboard with ML-based anomaly detection for live device activity visualization.	
Heavy Equipment Damage Detection (AI/ML - Computer Vision)	5 Months
· Developed AI/ML pipeline using ResNet-50 and YOLOv8 for damage assessment, achieving 93% accuracy across 5 categories. · Engineered event-driven validation layer using AWS Lambda + SQS, processing 2K+ images/day from Snowflake and S3. · Implemented config-based architecture enabling reusable data ingestion across 12+ data sources.	
LLM-Powered Salesforce Marketing Cloud Integration	5 Months
· Developed scalable integration platform using Python, AWS services, and LLM-assisted content generation for SFMC. · Implemented SOAP/REST API integrations with AI-powered campaign optimization, automating sync for 500K+ customer records. · Built monitoring with CloudWatch for error handling, retries, and model performance, achieving 99.5% uptime.	
AI Agent for Automated Code Review & Documentation	3 Months
· Built autonomous AI agent using LangChain + GPT-4 for code review, doc generation, and tech debt identification. · Implemented tool-calling patterns for Git integration and static analysis, reviewing 200+ PRs with 88% acceptance rate. · Designed evaluation framework with custom metrics for code quality and hallucination-free documentation generation.	

EDUCATION

B.Tech / B.E. Computer Science and Engineering (CSE)	2026
Sagar Institute of Science Technology and Research, Bhopal Grade: 6.5/10	
12th - Madhya Pradesh 74% (2022) 10th - Madhya Pradesh 89% (2020)	

CERTIFICATIONS

LangChain & LLM Application Development | Azure AI Fundamentals (AI-900) | AWS Cloud Practitioner | Databricks Lakehouse

ACHIEVEMENTS

- Promoted to Data Team Lead at CloudNexus within first year for strong technical delivery and AI innovation.
- AI Hackathon Winner: Built LLM-powered data quality agent that reduced pipeline failures by 45%.

ADDITIONAL INFORMATION

Hobbies: Technical Blogging, Open Source Contribution, Running, AI Research | Languages: English, Hindi
LinkedIn: <https://linkedin.com/in/harshit-vanya> | GitHub: <https://github.com/Harshit-Vanya>